

Ensuring AI Safety by Understanding Behaviors and Avoiding Being Predicted by other AI



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Introduction

In competitive settings, being predictable leads to exploitation, so developing algorithms with elements of unpredictability ensures they are robust and harder to exploit. This principle has many applications in cybersecurity, where systems must withstand adversarial attacks by preventing attackers from predicting their behavior. We study this problem in the framework of *two-player zero-sum normal-form games*.

Our work builds on that of Dinh, Nguyen et al. [1], who introduced algorithms (AFTRL and Prod-BR) capable of exploiting no-regret learners. Their approach assumes that the learner has additional knowledge of the opponent's behavior and that the difference between consecutive opponent strategies is 'small enough' to allow prediction of the next strategy from the past one.

In this project, we aim to design learning algorithms that mitigate such exploitation by incorporating controlled unpredictability. Our objective is to balance the trade-off between unpredictability and worst-case performance, contributing to the broader study of adversarial learning in machine learning and game theory.

1) Background

Two-Player Zero-Sum Normal-Form Game

A two-player zero-sum normal-form game consists of two players $\{1, 2\}$, where each player i has n_i pure strategies. The game is represented by a pay-off matrix $A \in \mathbb{R}^{n_1 \times n_2}$, where A_{ij} denotes the payoff to Player 1 when they play pure strategy i and Player 2 plays pure strategy j . Player 1 chooses a mixed strategy $f \in \Delta_{n_1}$ and Player 2 chooses a mixed strategy $y \in \Delta_{n_2}$. The expected utility to Player 1 is given by

$$u_1(f, y) = f^\top A y,$$

and since the game is zero-sum, Player 2's utility is

$$u_2(f, y) = -u_1(f, y).$$

Hence the game is zero-sum:

$$u_1 + u_2 = 0$$

Repeated Games

The game is played for T iterations. At each time step $t \in [T] = \{1, \dots, T\}$, the players decide on a strategy based on previous games played.

Then the players observe a feedback vector from the game.

- Player 1 observes feedback: Ay
- Player 2 observes feedback: $-A^\top f$

Nash Equilibrium

We have a solution to the game when a Nash Equilibrium is found. In a two-player zero-sum game, The Nash Equilibrium is characterised by Von Neumann's **Minimax Theorem** [2]:

$$\max_{f \in \Delta^{n_1}} \min_{y \in \Delta^{n_2}} f^\top A y = \min_{y \in \Delta^{n_2}} \max_{f \in \Delta^{n_1}} f^\top A y = v,$$

where v is the value of the game.

The point (f^*, y^*) satisfying the above is the Nash Equilibrium.

No-Regret

No-regret algorithms guarantee a level of performance in the worst case and achieve *average convergence* to Nash Equilibrium under self-play. This makes no-regret algorithms an attractive choice to ensure robustness and performance.

2) External Regret

The regret of a player after T rounds is defined as:

$$R_T = \max_{f^* \in \mathcal{F}} \left\{ \sum_{t=1}^T u_t(f^*) \right\} - \sum_{t=1}^T u_t(f_t)$$

where $\mathcal{F}$ is the set of possible strategies, f_t is the action chosen at time t , and $u_t(f)$ is the utility if strategy f were chosen at t .

The best fixed action $f^* \in \mathcal{F}$ that maximises total utility over all T rounds.

A no-regret algorithm satisfies:

$$\lim_{T \rightarrow \infty} \frac{R_T}{T} = 0$$

3) Proposed Algorithm

No-Regret Algorithm with Trigger: Let $\mathcal{A}$ a base no-regret algorithm, and $[a, b] \subset \mathbb{R}$ a payoff range. Initialize strategy $f_1 \in \mathcal{F}$. Then for $t = 1, \dots, T$:

$$f_t = \begin{cases} \frac{1}{n} \mathbf{1} & \text{if } f_t \cdot x_t \in [a, b], \\ f_t & \text{otherwise} \end{cases}$$

Observe feedback vector $x_t \in [0, 1]^n$

$$f_{t+1} = \mathcal{A}(f_1, \dots, f_t; x_1, \dots, x_t)$$

Repeat for all $t = 1, \dots, T$.

Our algorithm above proposes an adaption to any no-regret learning algorithm whereby upon receiving a payoff within a certain range it will play a uniform random strategy.

Further extension to this algorithm could be adapted to put multiple payoff ranges of varying sizes, decaying the width of the payoff range over time or having a budget for number of triggers. We leave this to future work.

However, our algorithm playing against an adversary who knew the payoff range beforehand or was able to learn it is extremely vulnerable to manipulation. Such an adversary could strategically exploit the learning algorithm, forcing it to play uniformly random action whenever it benefited them. While this represents a fundamental weakness in the approach, such an adversary is rare in practice.

4) Algorithm Analysis

Both proofs for (AFTRL and Prod-BR) [1] achieving better performance rely on the assumption that the consecutive strategies of the no-regret opponent be small i.e. For any norm $\|\cdot\|_q$:

$$\|f_{t+1} - f_t\|_q = O\left(\frac{1}{\sqrt{T}}\right)$$

The proposed algorithm successfully 'breaks' the proofs because the feedback observed will not be correlated to the previous feedback. Therefore they lose their theoretical guarantee of better performance.

5) Results and Conclusion

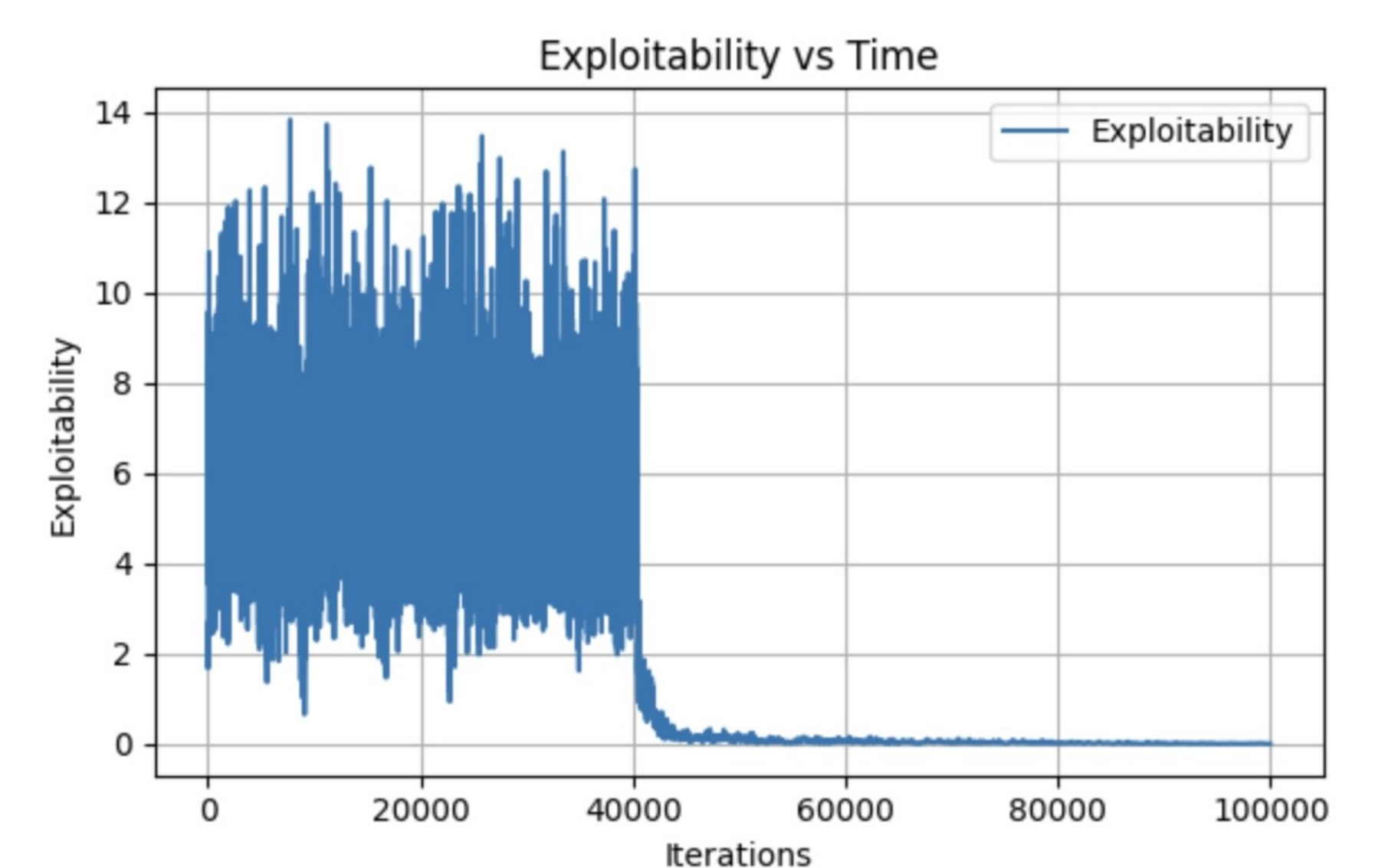


Figure 1: Exploitability plot.

We experimented on a 50×50 random matrix game using OMWU [3]. In self-play we discovered that our algorithms converges if the value of the game is not within the payoff range. From Figure 1 the algorithm can still converge to a Nash Equilibrium, which is seen at around iteration 40000, even if it starts by triggering.

Hence we show in this project it is possible to create an unpredictable algorithm which under certain conditions can achieve average convergence within self play and can successfully deceive smarter learning agents.

References

- (1) L. C. Dinh, T.-D. Nguyen et al., *Achieving Better Regret against Strategic Adversaries*, 2023.
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- (3) C. Daskalakis and I. Panageas, *arXiv preprint arXiv:1807.04252*, 2018.

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